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PROJECT AND TEAM INFORMATION

Project Title

Grade Grid – A comprehensive lab evaluation platform

Student/Team Information

Team Name:	FS-VI-T201-2026
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PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

In a typical academic environment, an educator often manages multiple sections, each consisting of approximately 70 to 80 students. This high volume of students makes the manual collection of lab assignments disorganized and increases the risk of work being misplaced. Furthermore, students often face a "transparency gap," where it is difficult to understand how their assignments were evaluated, the specific criteria used for grading, or the reasoning behind the marks and remarks provided by the teacher.

Grade Grid is a comprehensive lab evaluation platform designed to solve these logistical and educational challenges. By creating a full-stack website, the project aims to simplify the evaluation workflow for teachers while providing students with clear visibility into the basis of their grades. The platform allows teachers to set specific grading criteria and provide direct feedback, ensuring that students understand their performance and areas for improvement.

Updated Project Approach and Architecture (2 pts)

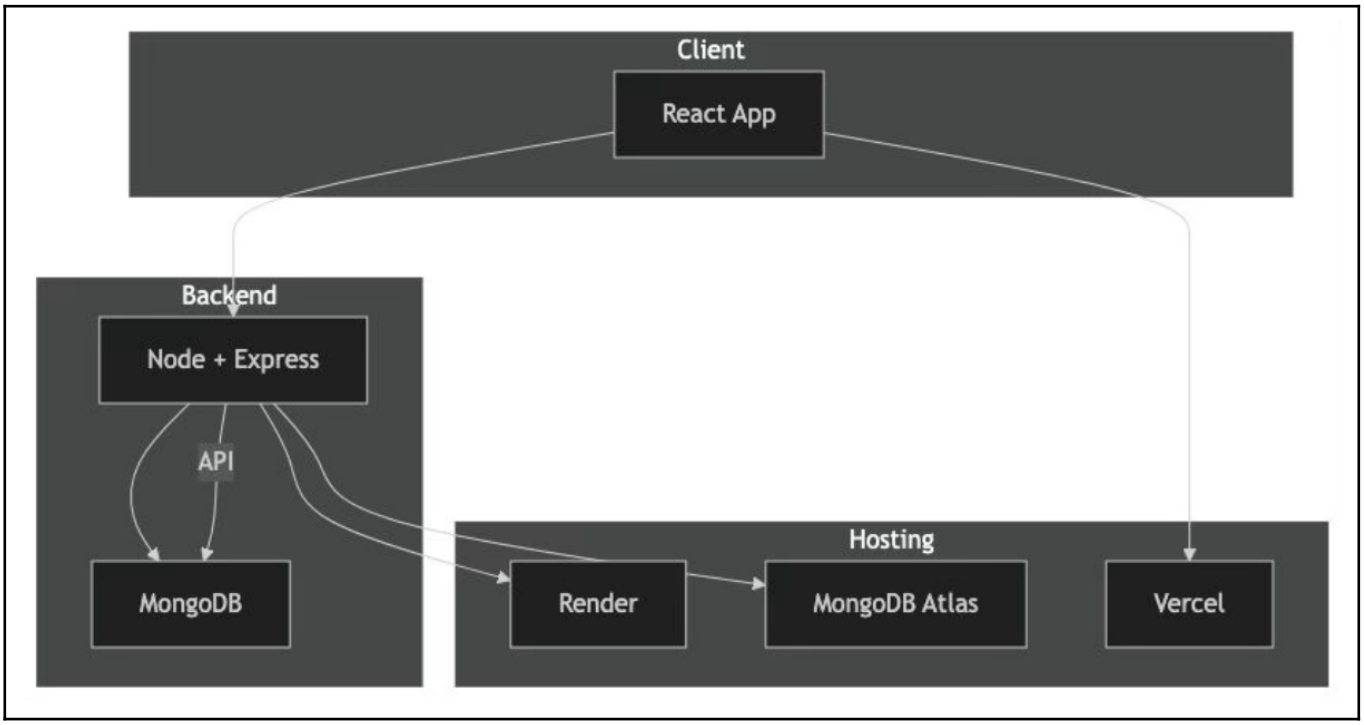
We are utilizing the MERN (MongoDB, Express.js, React, Node.js) stack to build a responsive, single-page application.

Frontend: Built with React to handle complex user interfaces, including a teacher dashboard and a multi-step assignment creation wizard.

Backend: A Node.js and Express server manages the RESTful API, handling authentication and data processing.

Database: MongoDB Atlas is used for its flexible NoSQL schema, allowing us to store diverse assignment metadata, student submissions, and rubric criteria.

Communication: The frontend communicates with the backend via API calls, ensuring data persistence for grades and remarks.



Tasks Completed (7 pts)

Task Completed	Team Member
Teacher Home Page: Designed the interface for teacher home page.	Achintya Bist
Assignment Creation: Built the UI for the page that allows teachers to create assignments with options to set criteria and due dates.	Priyanshu Sundriyal, Kartik Negi
Login Layouts: Created the frontend pages for user authentication for both roles.	Kartik Negi
Student Home Page: Designed the interface for student home page.	Achintya Bist, Amisha Bhandari
Navigation Bar UI: Developed the layout for navigation bar for both students and teachers.	Priyanshu Sundriyal
Navigation & Routing: Implemented client-side routing to navigate between different dashboard views.	Amisha Bhandari

Challenges/Roadblocks (7 pts)

The primary technical hurdle encountered during this phase was the development of the Multi-Step Rubric Builder. Implementing this feature in React required managing highly complex, nested states for dynamic criteria—such as category names, percentage weightages, and detailed descriptions—across several navigation steps. Because backend development has not yet begun, a significant challenge involves maintaining data integrity purely within the client-side state during a session. Ensuring that temporary data persists as a user moves back and forth through the wizard without a database for storage is a complex state-management task.

Additionally, designing a UI/UX that remains intuitive while offering such granular control has been a major roadblock. We have had to perform multiple design iterations to ensure the "Add Criteria" logic is flexible enough to handle varying lab assignment types without overwhelming the teacher with a cluttered interface. There is also the looming challenge of ensuring that the frontend data structure we are currently building perfectly aligns with our future MongoDB NoSQL schema. Without a backend in place to validate these structures, we risk significant refactoring once we begin the integration of Node.js and Express.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Backend Setup: Initializing the Node.js and Express.js server environment.	Amisha Bhandari
Database Integration: Setting up MongoDB Atlas and creating schemas for Users and Assignments.	Kartik Negi
API Development: Building RESTful endpoints to connect the React UI to the database.	Priyanshu Sundriyal
Authentication Logic: Implementing backend logic for secure login and role-based access.	Achintya Bist
Deployment: Hosting the final backend on Render.	Kartik Negi, Achintya Bist

Project Outcome/Deliverables (2 pts)

The primary deliverable of this project is a MERN-based web application designed to streamline lab assignment evaluation through a transparent, criteria-based feedback system.

Interactive Frontend Interface: A fully realized UI/UX built with React, featuring specialized dashboards for both teachers and students.

Assignment/Criteria Creation: The ability to create assignments with unique criterias for each assignment.

Documented Source Code: A complete, well-commented codebase for the entire application, managed and hosted in a public GitHub repository.

Data Export Utility: A feature allowing educators to export student marks and remarks into Excel or CSV files for administrative record-keeping.

The final product will serve as a centralized ecosystem that eliminates the "feedback gap" by providing students with clear justifications for their marks while reducing the administrative load on teachers.

Progress Overview (2 pts)

The project has successfully reached a significant milestone with the total completion of the UI/UX design and the high-fidelity interactive prototype of the website. This comprehensive phase involved building the entire visual framework for both students and teachers, including the intricate multi-step rubric builder and navigational dashboards, which puts the team ahead of schedule for all frontend-related milestones.

However, adhering to our specific development roadmap, no work has been initiated on the backend or database infrastructure yet. While the client-side interface is fully functional as a static prototype, the project is now in a transitional phase. Our primary focus is shifting entirely toward server-side logic, implementing the Node.js and Express.js environment, and establishing MongoDB connectivity. This transition will transform the current static UI into a dynamic, data-driven system capable of real-time evaluation and feedback.

Codebase Information (2 pts)

Repository link - https://github.com/kartik21777/Grade-Grid

Branch - Main

Initial commit: Grade Grid structure with Login and Dashboards - Created Login page and dashboards for teachers and students.

Created Card component - Created card component for reusability.

Finished UI design - Completed the UI with finishing touches.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
UI Component Test	Pass	All buttons, forms operate correctly within the React app.
Frontend Authentication UI	Pass	Successfully validated the login and signup page layouts for both student and teacher roles. The interface is specifically designed to authenticate users via a unique ID and Password rather than an email address.
Client-Side Navigation and Routing	Pass	Successfully validated that the application correctly navigates between the Login page, Teacher Dashboard, and Student Portal using React Router. All navigation links and buttons trigger the correct view transitions without requiring a page reload.

Deliverables Progress (2 pts)

MERN-based Web Application(In Progress): The frontend architecture is fully operational as a high-fidelity prototype; however, the full application remains in progress pending the backend and database integration phases.

Interactive Frontend Interface(Completed): The UI/UX for both teacher and student dashboards is 100% finalized using React, providing a modular and responsive foundation for the platform.

Assignment/Criteria Creation(Completed): The interactive interface for creating assignments and defining unique grading criteria is fully developed and functional within the frontend prototype.

Documented Source Code(In Progress): The frontend codebase is currently managed and well-documented in a public GitHub repository. Backend documentation will be added as development begins.

Data Export Utility(Pending): This feature is currently in the planning stage and is scheduled for implementation once the backend and file-generation logic are established.